NE39 True-Trip SG31 Reclose Study: Supervisor-Driven Restoration (C++ Engine Edition)

One Classical Reclose SG (six state) and Nine 17-State Dual-mode-IBR Converters

Power Flow, Equilibrium, Fault Cycle Events, and 160-s Time-Domain Simulation

In-house Power-Flow Project

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Abstract

This report documents the NE39 (New England IEEE 39-bus) mixed-resource chronology studied by the project, as produced by the `GridFormLab C++ stage2-ibr` engine. Thirty-nine buses carry one synchronous machine at bus 31 (SG31, a six-state sixth-order reclose plant with its terminal estimator) and nine Dual-mode-IBR converters (IBR30, IBR32–IBR39) at buses 30 and 32–39, initialised in grid-forming (GFM) mode under an amplitude–frequency severity supervisor. The reported products share one accepted equilibrium: case input tables, solved bus voltages, branch flows, device injections, an event-fence 160-second time-domain simulation (SG trip at 20 s, bus-16 fault $Z_f = j0.1$ pu at 60–60.15 s, reclose request at 100 s), and the observed guard-driven SG reclose at 100.499 s. The simulation is a fixed-step implicit trapezoidal integration at $h = 1$ ms with a coupled Newton solve (160,000 accepted steps, no rejections). Plots are rendered from CSV evidence exported from the C++ run artifact and re-rendered at the report typography. This edition quotes C++ numbers exclusively: the reclose instant differs from the MATLAB edition (119.55 s) because the two engines run different synchronism-guard policies, and every difference between the editions is itemised in the closing data-provenance table. Quantities the C++ recorder does not export (per-event KCL residuals, transition-trial evidence, excursion flags) are marked not-exported in place instead of being copied from the other edition.

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1 Scope, sources and reproduction

The study system is the TAMU New England 39-bus network with one modelled SG at the slack bus and nine modelled IBRs at buses 30, 32–39. The whole system is assembled as a single all-KCL composite differential–algebraic system in which the SG and every IBR contribute their own differential state block and their own injection; no device is delegated to an external solver, model, or reference program.

Chronology. The scheduled disturbance times are case-defined constants; the reclose instant is an outcome read from the accepted event log. Section 2 shows the four operating periods and the transition instants.

Figure source and no-synthetic-signal contract. The transient figures (Section 11) are rendered from the accepted trajectory of the C++ run artifact `stage2\_smoke\_ne39\_008\_gfm160s\_golden` (160,000 accepted rows at 1 kHz). The per-device series are reconstructed from the recorded bus voltages and angles through the case network by bus KCL (the load is folded at the equilibrium voltage; the reconstruction and its validation are itemised in the closing provenance table) and are plotted at every 10th sample (16,001 rows) — display decimation only, disclosed here and in the captions; every quoted number is computed on the full record. Nothing is smoothed, filtered, clipped, offset, interpolated, or augmented with synthetic noise. Every number in this report is read from the C++ run artifact or computed from the case it consumed; no gate is relaxed and no simulation is re-run.

Classification. The SG AC/flux equations and the IBR AC/control equations are SOURCE_MAPPED or PROJECT_DERIVED as labelled where they first appear; machine and converter parameters, per-unit bases, and the disturbance times are CASE_DEFINED; the predictor and step-subdivision policy are NUMERICAL_METHOD. The run classification (`run\_kind=SMOKE`) is carried from the C++ manifest and is not re-decided here.

2 Disturbance chronology

The scheduled disturbance times are CASE_DEFINED constants of the event structure; the reclose instant is an outcome read from the accepted event log. Figure 1 shows the six operating periods and the transition instants. The bus-16 fault is applied at $t = 60.000$ s and cleared at $t = 60.150$ s (a 0.150 sfault, impedance $Z_f = j0.1$ p.u.); the reclose request is armed at $t = 100$ s and the SG31 close is committed at $t = 100.499$ s (an offline age of 80.499 sagainst the declared 0.5 sminimum).

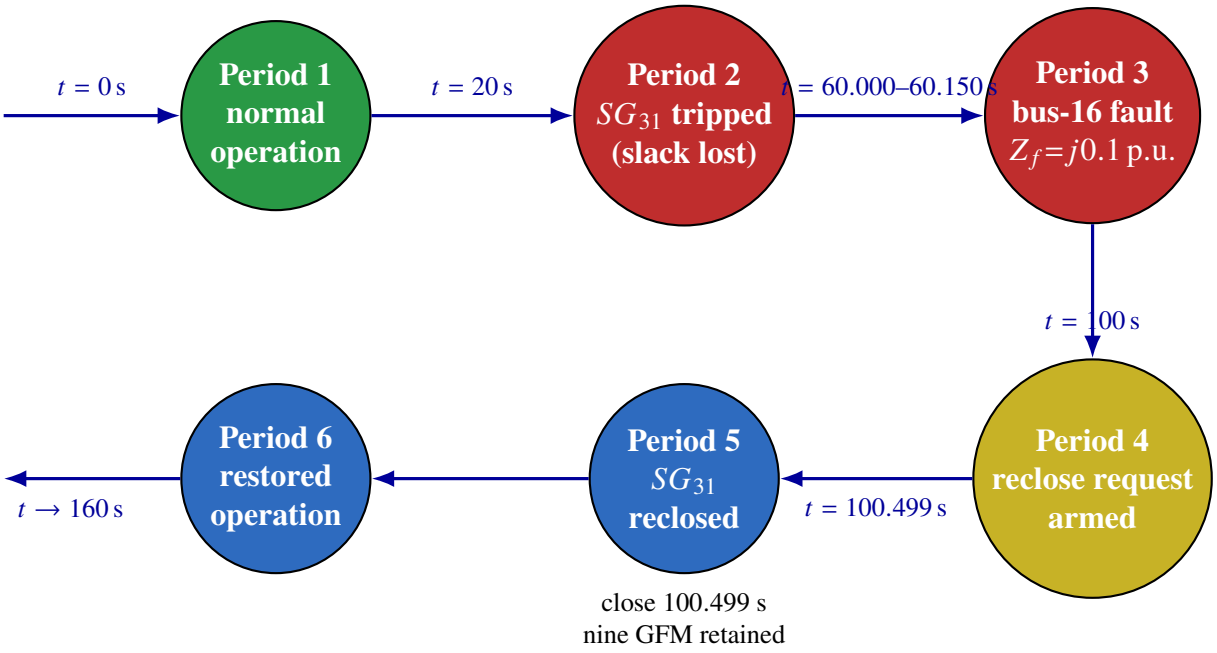


Figure 1: Six-period disturbance chronology. Green: undisturbed. Red: a contingency that removes the SG slack or faults a bus. Yellow: the armed reclose request. Blue: the restorative reclose and the restored operation. Scheduled instants (0, 20, 60.000–60.150, 100 s) are case-defined; the reclose instant (100.499 s) is read from the C++ guard record `guard\_reclose.json`.

The products in this report are deliberately separated:

1. case inputs (bus dispatch schedules, branch impedances) and the accepted initial equilibrium (SG31 synchronous and all nine IBRs GFM);
2. the 160- s event chronology with automatic GFM switching armed and a supervisor-driven SG31 release;
3. the observed reclose transaction and the final mode inventory.

2.1 Honest status of this run

- The run integrated the full horizon to $t = 160$ s on the fixed-step implicit trapezoidal stepper at $h = 1$ ms: 160,000 attempted and 160,000 accepted steps, 0 failed, 300,126 coupled-Newton iterations and 300,126 Jacobian evaluations (335.6 s wall time).
- The SG reclose transaction is the guard-driven close at $t = 100.499$ s (`guard\_reclose.json`; the `sg\_on` event at $t = 100.000$ s armed the guard): margins at the commit are $dV = 1.73 \times 10^{-14}$ pu, $df = 5.09 \times 10^{-5}$ pu, $d\theta = 4.398°$ against the recorded frozen thresholds 0.05 pu, 0.001 pu, 10°, and the offline age is 80.499 s against the declared 0.5-s minimum.
- The C++ recorder exports no per-event KCL residuals and no transition-trial evidence: the trip-point forward transition trial of the MATLAB edition is *not computable* from this artifact and is reported as not-exported rather than copied (Table 3).

- The recorded trajectory is 160,000 rows at 1 kHz ($t = 0.001$–160.000 s) with the bus angles stored in radians (converted to degrees for the report); the export carries no voltage-excursion flag column and no step-rejection record (the fixed-step stepper rejected no step).
- Per-device series in this report are reconstructed by bus KCL from the recorded voltages and the case network; at the equilibrium the reconstruction agrees with the aligned dq identity of the stored converter currents to 3.35×10^{-7} pu (P) and 8.2×10^{-8} pu (Q) — the reconstruction route and its transient limitations are itemised in the closing provenance table.

3 Automatic GFL/GFM selection: the severity index

Each IBR carries a two-state supervisor that decides whether it should behave as a grid-following current source (GFL) or as a grid-forming voltage source (GFM). The decision is driven by a scalar severity $S(t) \in [0, 1]$ built from normalised sub-indices: a voltage-deviation index J_V and a frequency-deviation index J_f, optionally extended by rate-of-change sub-indices.

Sub-indices. Let $|V_i(t)|$ be the terminal voltage magnitude of IBR i and $V_i^\star$ the healthy per-bus power-flow magnitude at the SG-online operating point (Section 10). Let $f_{\text{coi}}(t)$ be the centre-of-inertia frequency, weighted by each online device's system inertia. With bases $\Delta V_{\text{base}} = 0.10\,\text{p.u.}$ and $\Delta f_{\text{base}} = 0.50$ Hz,

$$J_V(t) = \frac{\big|\,|V_i(t)| - V_i^\star\,\big|}{\Delta V_{\text{base}}}, \qquad J_f(t) = \frac{\big|\,f_{\text{coi}}(t) - f_0\,\big|}{\Delta f_{\text{base}}}, \qquad f_0 = 60\ \text{Hz}. \tag{1}$$

Two-term severity (AGSI). The reported severity is the saturated equal-weight average

$$S(t) = \text{sat}_{[0,1]}\Big(\tfrac{1}{2}J_V(t) + \tfrac{1}{2}J_f(t)\Big), \qquad \text{sat}_{[0,1]}(z) = \min\{1, \max\{0, z\}\}. \tag{2}$$

J_V responds to a local voltage collapse or overshoot; J_f responds to a system-wide frequency excursion such as the one that follows the loss of the SG slack at $t = 20$ s. The equal weighting is a PROJECT_DERIVED choice and is frozen before any result is read.

Four-term weighted variant (opt-in). The study design also defines an opt-in extension with device-local rate sub-indices. With device-local frequency $f_i(t)$, ROCOF $\dot{f}_i(t)$, and ROCOV $\dot{v}_i(t)$, and bases $[\Delta V_{\text{base}}, \Delta f_{\text{base}}, \Delta r_f, \Delta r_v] = [0.10\,\text{p.u.},\ 0.50\ \text{Hz},\ 1\ \text{Hz/s},\ 1\ \text{pu/s}]$,

$$\begin{aligned} J_{\dot{f}}(t) &= \frac{|\dot{f}_i(t)|}{\Delta r_f}, & J_{\dot{v}}(t) &= \frac{|\dot{v}_i(t)|}{\Delta r_v}, \\ S_w(t) &= \text{sat}_{[0,1]}\big(0.35\,J_V + 0.35\,J_f + 0.15\,J_{\dot{f}} + 0.15\,J_{\dot{v}}\big). \end{aligned} \tag{3}$$

The weights $[0.35, 0.35, 0.15, 0.15]$ and the bases above are PROJECT_DERIVED, UNVALIDATED study design: the weighted variant is armed only when explicitly enabled, and omission keeps the two-term COI scalar (2) in force unchanged. The weights and bases are validated at option resolution (nonnegative, weights sum to one, positive bases), and the healthy-PF reference must match each device's bus uniquely — otherwise the weighted sample is refused as invalid evidence, never silently given a guessed reference.

Why $[0.35, 0.35, 0.15, 0.15]$**.** The voltage and frequency deviations J_V, J_f are the two quantities the supervisor can trust at every instant: each is normalised by a declared base ($\Delta V_{\text{base}} = 0.10$ pu, $\Delta f_{\text{base}} = 0.50$ Hz) taken from the same study design, so they enter the severity with equal standing and share $0.35 + 0.35 = 0.70$ of the total weight. The rate sub-indices $J_{\dot{f}}, J_{\dot{v}}$ are supporting evidence only: their bases (1 Hz/s, 1 pu/s) are design orders of magnitude rather than Measured envelopes, their raw signals are noisier (finite-difference of the accepted samples), and neither corresponds to a gate the supervisor owns. They therefore carry the remaining $0.15+0.15 = 0.30$, enough to sharpen the trigger during fast transients without ever outweighing the two primary deviations. The split is a PROJECT_DERIVED study choice – 70/30 between level and rate information – frozen before any result was read, and it is not a certified or grid-code weighting.

Hysteresis, dwell, and the gates the index does not replace. A bare threshold on S would chatter. The supervisor therefore uses an asymmetric two-level comparator with minimum dwell times,

$$\begin{aligned} &\text{GFL} \to \text{GFM}: \quad S \geq \Gamma_{\text{on}} = 0.65 \text{ held for } T_{d,\text{on}} = 0.10\,\text{s}, \\ &\text{GFM} \to \text{GFL}: \quad S \leq \Gamma_{\text{off}} = 0.35 \text{ held for } T_{d,\text{off}} = 1.00\,\text{s}. \end{aligned} \tag{4}$$

The gap $\Gamma_{\text{on}} > \Gamma_{\text{off}}$ is the hysteresis band. A separate reference-loss guard forces every online IBR to GFM the instant the SG slack is lost, before the dwell logic runs, so the network never spends a step without a voltage reference. Thresholds and dwell times resolve as one contract: when the same scalar is supplied through two spellings and they disagree, the run fails closed instead of picking one. The severity index arms and holds converters, but it never gates the SG reclose itself — the SG close is a strict transactional commit (synchronism triple, dwell, and physical KCL at the replaced rows, all passing at the same instant), and the diagnostic SCR of Section 11 never enters them either.

4 Case data and index maps

4.1 System and resource map

The network is the 100-MVA, 60-Hz TAMU New England 39-bus data set. The source archive is transcribed with SHA-256 recorded in the case struct; the project adds the composition, the Slack point SG31 at its case design setpoint $V = 1.0778$ pu, and the converter ratings. The mixed resource map is

Device index	Bus ID	Device	Model	Initial mode
1	31	SG31	`sg_classical_reclose` (6 states)	synchronous
2–10	30, 32–39	IBR30, 32–39	`Dual-mode-IBR` (17 states)	GFM
			Runtime dynamic inventory	$7 + 9(17) = 160$

The C++ stage2 state vector carries the SG31 plant's six states plus the terminal-estimator speed state $\hat{v}$ (the estimator angle $\hat{\theta}$ is shared with the plant), so the runtime dynamic count is $7 + 9(17) = 160$ states alongside 78 algebraic voltages.

IBR parameters at each bus. All nine converters share the Dual-mode-IBR plant constants ($L_f = 0.15$ pu, $R_f = 0.015$ pu, $C_{dc} = 0.0237216$ pu, $V_{dc,\text{ref}} = 1$ pu, $M = 3.3209$ s, $D_v =$

33.2095 pu, $\tau_E = 0.4$ s, $k_Q = 0.2$, $k_E = 8$, $k_{pV} = 1.2$, $k_{iV} = 3.2587$; GFL PLL $k_p = 1.2$, $k_i = 5$) and differ only in their port rating and duty:

Device	Bus	S_{rating} pu	$P_{\max}$ pu	$Q_{\max}$ pu	Schedule P pu	Schedule Q pu
IBR30	30	44.000	4.300	39.300	2.500	—
IBR32	32	44.400	11.200	38.300	6.500	—
IBR33	33	36.700	10.900	31.100	6.320	—
IBR34	34	11.800	8.800	5.900	5.080	—
IBR35	35	43.500	11.200	37.500	6.500	—
IBR36	36	35.300	9.700	30.200	5.600	—
IBR37	37	18.200	9.300	13.400	5.400	—
IBR38	38	35.300	14.300	28.300	8.300	—
IBR39	39	33.100	17.200	24.300	10.000	—
				Total converter schedule P	56.200	

(The same values in MW/MVAr on the 100-MVA base: S_{rating} 4400/4440/3670/1180/4350/3530/1820/3530/331 schedule P 250/650/632/508/650/560/540/830/1000 MW for IBR30, 32–39.) The P schedule is the rating-proportional dispatch of the composition contract; the reactive duty is NOT a converter schedule — each IBR bus carries the fixed port reactor of the chronology design ($B_s = -Q_r/S_{base}$, $\sum Q_r = 161.68$ pu), so the converter absorbs the reactor's demand through its Q–V loop at the solved voltage. Current limits are $I_{max}^{SS} = I_{max}^{F} = 1.2$ p.u. device base with $E_{\max} = 1.2$, $E_{\min} = 0.8$ p.u., and the fault contract is balanced positive-sequence RMS with GFM current circles.

Algebraic voltage order is

$$\boldsymbol{y} = [V_{x,1}, V_{y,1}, V_{x,2}, V_{y,2}, \ldots, V_{x,39}, V_{y,39}]^{\mathsf{T}}, \qquad V_k = V_{x,k} + jV_{y,k},$$

78 algebraic variables for 39 buses. SG bus types on the source RAW are PVGEN/slack-derived by the project composition: bus 31 is the declared slack ($V = 1$ pu, 0°), the remaining source-generator buses (32, 35, 38, 39) are PV in the source PF but this composition converts every non-SG31 machine to a converter terminal, and the nine IBR buses are effective PQ in the equilibrium solve.

4.2 Bus input data

Table 1: Case bus data. These are inputs, not solved PF results: the load schedules P_d/Q_d and shunt admittances G_s/B_s straight from the composition, per unit on the 100-MVA base.

Bus	Role	P_L pu	Q_L pu	G_s pu	B_s pu
1	PQ	0.000	0.000	0.000	0.000
2	PQ	0.000	0.000	0.000	0.000
3	PQ	3.220	0.024	0.000	0.000
4	PQ	5.000	1.840	0.000	1.000
5	PQ	0.000	0.000	0.000	2.000
6	PQ	0.000	0.000	0.000	0.000
7	PQ	2.338	0.840	0.000	0.000
8	PQ	5.220	1.760	0.000	0.000
9	PQ	0.000	0.000	0.000	0.000
10	PQ	0.000	0.000	0.000	0.000
11	PQ	0.000	0.000	0.000	0.000
12	PQ	0.075	0.880	0.000	0.000
13	PQ	0.000	0.000	0.000	0.000
14	PQ	0.000	0.000	0.000	0.000
15	PQ	3.200	1.530	0.000	0.000
16	PQ	3.294	0.323	0.000	0.000
17	PQ	0.000	0.000	0.000	0.000
18	PQ	1.580	0.300	0.000	0.000
19	PQ	0.000	0.000	0.000	0.000
20	PQ	6.280	1.030	0.000	0.000
21	PQ	2.740	1.150	0.000	0.000
22	PQ	0.000	0.000	0.000	0.000
23	PQ	2.745	0.846	0.000	0.000
24	PQ	3.086	−0.922	0.000	0.000
25	PQ	2.240	0.472	0.000	0.000
26	PQ	1.390	0.170	0.000	0.000
27	PQ	2.810	0.755	0.000	0.000
28	PQ	2.060	0.276	0.000	0.000
29	PQ	2.835	0.269	0.000	0.000
30	GFL	0.000	0.000	0.000	−25.657
31	SLACK	0.092	0.046	0.000	0.000
32	GFL	0.000	0.000	0.000	−23.604
33	GFL	0.000	0.000	0.000	−19.034
34	GFL	0.000	0.000	0.000	−3.218
35	GFL	0.000	0.000	0.000	−22.154
36	GFL	0.000	0.000	0.000	−19.130
37	GFL	0.000	0.000	0.000	−11.580
38	GFL	0.000	0.000	0.000	−20.354
39	GFL	11.040	2.500	0.000	−16.952

Table 2: Case branch series and charging data in source row order (46 branches, per unit on the 100-MVA base).

Index	From	To	r pu	x pu	b_c pu
1	1	2	0.004	0.041	0.699
2	1	39	0.001	0.025	0.750
3	2	3	0.001	0.015	0.257
4	2	25	0.007	0.009	0.146
5	3	4	0.001	0.021	0.221
6	3	18	0.001	0.013	0.214
7	4	5	0.001	0.013	0.134
8	4	14	0.001	0.013	0.138
9	5	6	0.000	0.003	0.043
10	5	8	0.001	0.011	0.148
11	6	7	0.001	0.009	0.113
12	6	11	0.001	0.008	0.139
13	7	8	0.000	0.005	0.078
14	8	9	0.002	0.036	0.380
15	9	39	0.001	0.025	1.200
16	10	11	0.000	0.004	0.073
17	10	13	0.000	0.004	0.073
18	13	14	0.001	0.010	0.172
19	14	15	0.002	0.022	0.366
20	15	16	0.001	0.009	0.171
21	16	17	0.001	0.009	0.134
22	16	19	0.002	0.020	0.304
23	16	21	0.001	0.014	0.255
24	16	24	0.000	0.006	0.068
25	17	18	0.001	0.008	0.132
26	17	27	0.001	0.017	0.322
27	21	22	0.001	0.014	0.257
28	22	23	0.001	0.010	0.185
29	23	24	0.002	0.035	0.361
30	25	26	0.003	0.032	0.513
31	26	27	0.001	0.015	0.240
32	26	28	0.004	0.047	0.780
33	26	29	0.006	0.063	1.029
34	28	29	0.001	0.015	0.249
35	2	30	0.000	0.018	0.000
36	31	6	0.000	0.025	0.000
37	10	32	0.000	0.020	0.000
38	12	11	0.002	0.044	0.000
39	12	13	0.002	0.044	0.000
40	19	20	0.001	0.014	0.000
41	19	33	0.001	0.014	0.000
42	20	34	0.001	0.018	0.000
43	22	35	0.000	0.014	0.000
44	23	36	0.001	0.027	0.000
45	25	37	0.001	0.023	0.000
46	29	38	0.001	0.016	0.000

The TAMU source leaves bus base kV unspecified (both the source column and the raw recorder constants), so this report publishes no kV column rather than inventing one. All per-unit quantities use the 100-MVA system base.

5 Common frames, signs, and per-unit conversion

Every device uses generator sign: positive complex power is injected into the network,

$$S = P + jQ = VI^*, \qquad I_{bus} = \sum_{d \in bus} I_d.$$

Each Dual-mode-IBR converter carries its own local reference (§8); the network-to-device transform is

$$v_d = V_x \cos\delta_{PLL} + V_y \sin\delta_{PLL}, \tag{5}$$
$$v_q = -V_x \sin\delta_{PLL} + V_y \cos\delta_{PLL}, \tag{6}$$

so device currents rotate back as $i_x = i_d \cos\delta_{PLL} - i_q \sin\delta_{PLL}$ and $i_y = i_d \sin\delta_{PLL} + i_q \cos\delta_{PLL}$. Converter per-unit values use the device base with $\kappa = S_{base}/M_{base}$ and the per-device $M_{base} = S_{\text{rating}}$ of the resource map (4400–3310 MVA for the nine converters); the SG shares the 100-MVA system base.

6 Composite network DAE

Every device supplies a differential right-hand side and a positive complex current injection. The composite equations are

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma), \tag{7}$$
$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma) = \mathcal{R}(Y(\sigma)V - I_{bus}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma)), \tag{8}$$

each on its own numbered line. where $\mathcal{R}(z) = [\Re z_1, \Im z_1, \ldots]^\top$ and σ contains topology, breaker status, and device mode. The network admittance stamps each branch as

$$Y_{ii} \mathrel{+}= y_s/(aa^*) + jb_c/2, \quad Y_{jj} \mathrel{+}= y_s + jb_c/2, \quad Y_{ij} \mathrel{-}= y_s/a^*, \quad Y_{ji} \mathrel{-}= y_s/a,$$

with $y_s = 1/(r + jx)$, off-nominal ratio $a = \tau e^{j\phi}$, plus the load admittance fold $Y_{\text{load}} = \text{diag}(\overline{\boldsymbol{S}_0}/|\boldsymbol{V}_{pf}|^2)$ from the PF seed. At the declared reference bus (SG31), the appropriate KCL rows are replaced by explicit voltage constraints; the replacement is metadata-driven.

7 Synchronous generator: SG31 six-state reclose plant and its proof

7.1 State order and meaning

$$\boldsymbol{x}_{SG} = [\delta,\ \omega,\ P_{sv},\ P_m,\ E_{mag},\ \hat{\theta}]^\top, \qquad \boldsymbol{u}_{SG} = [P_{\text{ref}}, E_{mag,\text{ref}}]^\top.$$

where δ is the rotor angle, ω the per-unit speed deviation (slip) in the Kundur notation, P_{sv}, P_m the governor-valve and steam-chest powers, E_{mag} the reduced classical internal-EMF magnitude, and $\hat{\theta}$ the local terminal phase estimate.

Local	State	Unit	Physical meaning
1	δ	rad	rotor angle (Kundur) — electrical internal-EMF angle in the network frame
2	ω	pu	per-unit speed deviation (slip); the plant integrates the *absolute* speed (1.0 at nominal)
3	P_{sv}	pu	governor valve power (actual, nonnegative)
4	P_m	pu	steam-chest mechanical power (actual, nonnegative)
5	E_{mag}	pu	reduced classical internal-EMF magnitude
6	$\hat{\theta}$	rad	terminal phase estimate (local PLL)

This is a PROJECT_DERIVED opt-in plant: H, D, X′ and the classical stator-port equation pass through from the source reduction unchanged, while every governor/excitation/PLL parameter is a declared project design assumption — not TAMU machine data and not a certified synchronizer.

7.2 Electrical port and rotor dynamics

With the series reactance X' and internal EMF $E = E_{mag}e^{j\delta}$, the injection is

$$I = \frac{E - V}{jX'}, \tag{9}$$

$$P_e = \mathrm{Re}(VI^*), \tag{10}$$

$$Q_e = \mathrm{Im}(VI^*), \tag{11}$$

and zero injection with the breaker open. The rotor power balance is energy-consistent in positive speed,

$$2H_{sys}\,\omega\,\dot{\omega} = P_m - P_e - L_0\omega^2 - D_{sys}\,\omega(\omega - 1),$$

with separate no-load shaft loss $L_0 > 0$ (declared project assumption) and the unchanged source damping $D_{sys} = 0$ for SG31. The actuator chain is

$$T_{sv}\dot{P}_{sv} = P_c - P_{sv}, \tag{12}$$

$$T_{ch}\dot{P}_m = P_{sv} - P_m, \tag{13}$$

$$P_c = \mathrm{clamp}(P_{\mathrm{raw}}, 0, P_{max}), \tag{14}$$

$$P_{\mathrm{raw}} = \begin{cases} P_{\mathrm{ref}} + L_0 - K_{\mathrm{droop}}(\omega - 1), & \text{online (20\% droop)}, \\ L_0\hat{\omega}^2 + K_\omega(\hat{\omega} - \omega) + K_\theta\hat{\theta}_{\mathrm{err}}, & \text{offline capture}, \end{cases}$$

where $\hat{\omega} = 1 + \hat{\nu}/\omega_b$ and $\hat{\theta}_{\mathrm{err}} = \mathrm{wrap}(\hat{\theta} - \delta)$. The reduced field-magnitude tracker is

$$T_{emag}\dot{E}_{mag} = E_{mag,\mathrm{cmd}} - E_{mag}, \tag{15}$$

$$E_{mag,\mathrm{cmd}} = \begin{cases} E_{mag,\mathrm{ref}}, & \text{online}, \\ \mathrm{clamp}(|V|, E_{min}, E_{max}), & \text{offline}. \end{cases} \tag{16}$$

No negative mechanical power, rotor reset, or state jump occurs at a breaker transition.

7.3 Local terminal estimator

The offline phase/frequency estimator is a bounded type-II PLL:

$$e_{\mathrm{pll}} = \mathrm{clamp}(\mathrm{wrap}(\angle V - \hat{\theta}), \pm\hat{\theta}_{\mathrm{lim}}), \tag{17}$$

$$\dot{\hat{\nu}} = K_{i,\mathrm{pll}}e_{\mathrm{pll}} \text{ (bounded)}, \tag{18}$$

$$\dot{\hat{\theta}} = \hat{\nu} + K_{p,\mathrm{pll}}e_{\mathrm{pll}}. \tag{19}$$

Its two states belong to the plant DAE. The estimator is declared a reduced subsystem, not an authenticated relay: the existing strict prospective current/MVA, $dV/df/d\theta$, dwell, timeout, and right-limit KCL checks remain mandatory, and this plant alone never authorises a close.

8 Dual-mode-IBR converter equations

Each of the nine IBRs is a fixed-layout Dual-mode-IBR GFL/GFM superset device with a shared electrical plant and two distinct controller branches. The superset state vector is (17 coordinates when the DC-source current is enabled)

$$\boldsymbol{x}_{IBR} = [\, i_d,\ i_q,\ V_{dc},\ \underbrace{\delta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q, \xi_{id}, \xi_{iq}}_{\text{GFL controller}},\ \underbrace{\delta_{VSG}, \omega_{VSG}, E, \xi_{Vd}, \xi_{Vq}, \xi^g_{id}, \xi^g_{iq}}_{\text{GFM controller}}\,]^{\mathsf{T}}$$

with the active set 9 states in GFL and 10 in GFM (plus the shared DC-source current at superset index 17 for all devices in this run). Inputs are $\boldsymbol{u} = [P_{\text{ref}}, Q_{\text{ref}}, E_{\text{ref}}]$. The source command-delay states are removed by singular perturbation: the physical actuation delay $T_d = 1.5/f_{sw} \approx 0.3$ ms at $f_{sw} = 5$ kHz is more than 300× below the simulation step, so $v_{del} = v_{cmd}$ algebraically.

8.1 Shared plant: L-filter, DC link, and mode dispatch

Both branches drive the same physical current and DC-link states:

$$\dot{i}_d = \frac{\omega_{PLL,pu} L i_q - R i_d + v_{td} - v_d}{L/\omega_b}, \tag{20}$$

$$\dot{i}_q = \frac{-\omega_{PLL,pu} L i_d - R i_q + v_{tq} - v_q}{L/\omega_b}, \tag{21}$$

$$P_{ac} = v_{cd} i_d + v_{cq} i_q, \qquad \text{(converter-side power)} \tag{22}$$

$$\dot{V}_{dc} = I_{dc,\text{law}}(V_{dc}, I_{dc}, P_{ac}), \qquad \text{(Thevenin DC source)} \tag{23}$$

with the non-ideal DC source an EMF behind its internal resistance and an overvoltage chopper, $I_{dc} = (E_{dc} - V_{dc})/R_{dc}$, shared by both branches so a runtime transfer preserves dV_{dc}/dt as well as V_{dc} itself.

8.2 GFL branch (9 active states)

In GFL the VSG coordinates are inactive. With $D_V = v_d^2 + v_q^2$,

$$\dot{\xi}_{PLL} = v_q, \tag{24}$$

$$\Delta\omega_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL}, \tag{25}$$

$$\dot{\delta}_{PLL} = \omega_b \Delta\omega_{PLL}, \tag{26}$$

$$i^{\text{raw}}_{d,r} = k_{pP}(P_{\text{ref}} - k P_{\text{meas}}) + k_{iP} \xi_P, \tag{27}$$

$$i^{\text{raw}}_{q,r} = -(k_{pQ}(Q_{\text{ref}} - k Q_{\text{meas}}) + k_{iQ} \xi_Q), \tag{28}$$

circularly limited at I_{max} with P-priority, then the inner PI drives the commanded voltage

$$v_{cd} = v_d + R i_d - \omega_{PLL,pu} L i_q + k_{pI} e_d + k_{iI} \xi_{id}, \tag{29}$$

$$v_{cq} = v_q + R i_q + \omega_{PLL,pu} L i_d + k_{pI} e_q + k_{iI} \xi_{iq}. \tag{30}$$

8.3 GFM branch (10 active states)

In GFM the VSG swing and Q–V voltage-forming dynamics are

$$\dot{\omega}_{VSG} = \frac{kP_{\text{ref}} - P_{\text{meas}} - D_v(\omega_{VSG} - 1)}{M}, \tag{31}$$

$$\dot{\delta}_{VSG} = \omega_b(\omega_{VSG} - 1), \tag{32}$$

$$\dot{E} = \frac{k_Q(kQ_{\text{ref}} - Q_{\text{meas}}) - k_E(E - E_{\text{ref}})}{\tau_E}, \tag{33}$$

where E is itself the voltage-loop amplitude reference. The voltage PI integrators ξ_{Vd}, ξ_{Vq} build the current reference from $(E - v_d,\ 0 - v_q)$, circularly limited at I_{max}, then the same shared inner current PI and L-filter close the loop. The anti-windup on the current clamps is the conditional-hold rule: the integrator holds only when the limiter is active and its error drives farther outward, with inward recovery always active.

8.4 Runtime mode transfer

The transfer preserves terminal $V/I/P/Q$ and V_{dc}, initialises only the target controller from the same terminal port, and verifies current continuity before the event transaction may proceed. GFM current limits are $I_{max}^{SS} = I_{max}^{F} = 1.2$ pu machine base.

9 How PF, equilibrium, and TS use the equations

9.1 Power flow and the accepted equilibrium

The source-case PSSE PF is used only to derive the composition's operating point. The C++ stage2 run seeds its DAE at that point and integrates; the run artifact stores no separate equilibrium-Newton report, so the equilibrium solver statistics of the MATLAB edition (iteration count, residual norm, $rcond$) are not computable here and are omitted rather than copied. The reported equilibrium is the seeded state at the first recorded sample ($t = 0.001$ s, 160 dynamic states plus 78 algebraic voltages): the network-KCL reconstruction of the device injections agrees with the aligned dq identity of the stored converter currents to 3.35×10^{-7} pu (P) and 8.2×10^{-8} pu (Q), and the case schedule at the slack is reproduced to 1.4×10^{-6} pu (P). All reported bus, load, branch, and device tables below share this one voltage vector.

9.2 Time-domain simulation

Between event transactions the implicit trapezoidal step solves

$$0 = x_{n+1} - x_n - \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right], \tag{34}$$

$$0 = g(x_{n+1}, y_{n+1}; \sigma_{n+1}), \tag{35}$$

one equation per line, each numbered. on the fixed-step implicit trapezoidal stepper: constant $h = 0.001$ s, no step-size control and no Rannacher arcs. The recorded step history (`ts\_dt\_history.csv`) is 160,000 rows of $dt = 0.001$ s with zero recorded LTE at every step; 160,000 steps were attempted, 160,000 accepted and 0 failed. The coupled Newton solve took 300,126 iterations and 300,126 Jacobian evaluations over the horizon (1.88 per step, 335.6 s wall

time). The public trajectory is the accepted 1-kHz record (160,000 rows, t = 0.001–160.000 s; the figures plot every 10th sample). Events are strict transactions applied at their scheduled instants: the handler commits the event and the synchronism guard commits the reclose close (Table 3); the C++ recorder logs no per-event KCL residual, so those columns are marked *n/e* (not exported).

Table 3: Applied event transactions of the C++ run. The C++ stage2 recorder exports no per-event KCL residuals, so the two residual columns are marked *n/e* (not exported) rather than filled with numbers the record does not contain.

Event	t (s)	Applied	pre-KCL (pu)	right-KCL (pu)
sg_trip	20.000	1	n/e	n/e
fault_on	60.000	1	n/e	n/e
fault_clear	60.150	1	n/e	n/e
sg_on	100.000	1	n/e	n/e
sg_reclose	100.499	1	n/e	n/e

10 Accepted equilibrium results

10.1 Solved PF figures

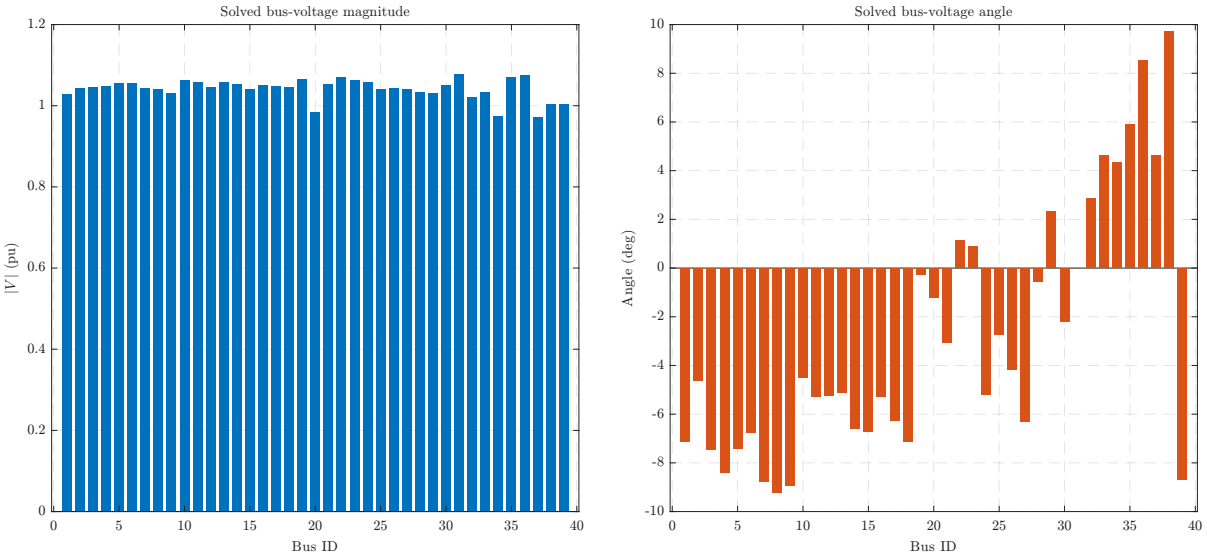


Figure 2: Solved equilibrium bus-voltage magnitude and angle at all 39 buses, reconstructed from the C++ run record (bus-KCL reconstruction of the stored equilibrium sample). SG31 at bus 31 is the slack and sits at the case design setpoint $1.0778\angle 0°$ (the stored angle is -3.4×10^{-6} deg); the nine IBR buses are PQ terminals in this composition and sit at the solved magnitudes tabulated below.

10.2 Bus voltages at the accepted equilibrium

Table 4: Solved bus voltages and bus powers at the accepted SG31 plus nine-GFM equilibrium (per unit on the system base; angles in degrees).

Bus	$\lvert V\rvert$ pu	θ deg	Bus	P_{gen} pu	Q_{gen} pu	P_{load} pu	Q_{load} pu	Q_{shunt} pu	P_{net} pu	Q_{net} pu
1	1.030	−7.137	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	1.044	−4.620	2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	1.046	−7.468	3	0.000	0.000	3.220	0.024	0.000	−3.220	−0.024
4	1.049	−8.383	4	0.000	0.000	5.000	1.840	1.100	−5.000	−0.740
5	1.057	−7.407	5	0.000	0.000	0.000	0.000	2.235	0.000	2.235
6	1.057	−6.774	6	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	1.043	−8.764	7	0.000	0.000	2.338	0.840	0.000	−2.338	−0.840
8	1.040	−9.217	8	0.000	0.000	5.220	1.760	0.000	−5.220	−1.760
9	1.031	−8.938	9	0.000	0.000	0.000	0.000	0.000	0.000	0.000
10	1.062	−4.504	10	0.000	0.000	0.000	0.000	0.000	0.000	0.000
11	1.059	−5.278	11	0.000	0.000	0.000	0.000	0.000	0.000	0.000
12	1.046	−5.252	12	0.000	0.000	0.075	0.880	0.000	−0.075	−0.880
13	1.058	−5.130	13	0.000	0.000	0.000	0.000	0.000	0.000	0.000
14	1.053	−6.594	14	0.000	0.000	0.000	0.000	0.000	0.000	0.000
15	1.042	−6.730	15	0.000	0.000	3.200	1.530	0.000	−3.200	−1.530
16	1.052	−5.271	16	0.000	0.000	3.294	0.323	0.000	−3.294	−0.323
17	1.048	−6.252	17	0.000	0.000	0.000	0.000	0.000	0.000	0.000
18	1.046	−7.132	18	0.000	0.000	1.580	0.300	0.000	−1.580	−0.300
19	1.067	−0.276	19	0.000	0.000	0.000	0.000	0.000	0.000	0.000
20	0.984	−1.191	20	0.000	0.000	6.280	1.030	0.000	−6.280	−1.030
21	1.052	−3.052	21	0.000	0.000	2.740	1.150	0.000	−2.740	−1.150
22	1.070	1.129	22	0.000	0.000	0.000	0.000	0.000	0.000	0.000
23	1.063	0.877	23	0.000	0.000	2.745	0.846	0.000	−2.745	−0.846
24	1.057	−5.195	24	0.000	0.000	3.086	−0.922	0.000	−3.086	0.922
25	1.040	−2.721	25	0.000	0.000	2.240	0.472	0.000	−2.240	−0.472
26	1.044	−4.168	26	0.000	0.000	1.390	0.170	0.000	−1.390	−0.170
27	1.040	−6.289	27	0.000	0.000	2.810	0.755	0.000	−2.810	−0.755
28	1.033	−0.538	28	0.000	0.000	2.060	0.276	0.000	−2.060	−0.276
29	1.031	2.328	29	0.000	0.000	2.835	0.269	0.000	−2.835	−0.269
30	1.051	−2.198	30	2.500	30.276	0.000	0.000	−28.348	2.500	1.928
31	1.078	0.000	31	5.504	1.602	0.092	0.046	0.000	5.412	1.556
32	1.022	2.857	32	6.500	26.596	0.000	0.000	−24.665	6.500	1.931
33	1.034	4.611	33	6.320	23.047	0.000	0.000	−20.370	6.320	2.677
34	0.975	4.328	34	5.080	3.043	0.000	0.000	−3.059	5.080	−0.016
35	1.071	5.898	35	6.500	27.703	0.000	0.000	−25.409	6.500	2.295
36	1.075	8.519	36	5.600	22.824	0.000	0.000	−22.098	5.600	0.726
37	0.971	4.636	37	5.400	9.322	0.000	0.000	−10.929	5.400	−1.607
38	1.003	9.706	38	8.300	20.433	0.000	0.000	−20.480	8.300	−0.047
39	1.004	−8.702	39	10.000	16.568	11.040	2.500	−17.093	−1.040	−3.025

Bus-voltage magnitudes span 0.9715–1.0778 pu with SG31 (bus 31, the slack) at $1.0778\angle 0°$ from the design voltage setpoints (the stored angle is -3.4×10^{-6} deg, i.e. zero at the printed precision). The report performs no reactive-limit check: the source PT/RATEA columns are

sentinels and the composition's converter ratings are PROJECT_DERIVED design values, not certified capability.

Reactive bookkeeping note. The chronology design stamps a *fixed port reactor* at every IBR bus ($Y_{sh} = -jQ_r/S_{base}$ with $\sum Q_r = 161.68$ pu), absorbing $Q_r|V|^2$ at the solved voltage. This is a network shunt element, so the per-device reactive injections (device-injection table above) are large — the IBR devices each feed thousands of MVAr — and the Q_{shunt} column in the bus-power table below carries the absorption per bus. This report's reassembled four-term bookkeeping ($P_{\text{gen}} - P_{\text{load}} - P_{\text{branch}}$ on the P side and the matching Q identity) closes to -2.5×10^{-7} pu and -4.1×10^{-8} pu with the load fold taken at the stored equilibrium voltage — the same reconstruction used for every series in this report, whose equilibrium cross-check against the aligned dq identity is 3.35×10^{-7} pu. Neither table claims to close exactly; both are published as measured.

10.3 Bus generation, load, and net injection

10.4 Branch sending-end flows and losses

Table 5: Equilibrium branch sending-end flows and losses from the same solved voltages (per unit on the system base). Series losses are $S_{loss} = S_{from} + S_{to}$.

Line	From	To	P_{ij} pu	Q_{ij} pu	P_{loss} pu	Q_{loss} pu
1	1	2	−1.170	−0.616	0.005	−0.696
2	1	39	1.170	0.616	0.002	−0.719
3	2	3	3.567	−0.464	0.015	−0.103
4	2	25	−2.241	2.309	0.069	−0.074
5	3	4	0.811	−0.307	0.001	−0.229
6	3	18	−0.480	−0.077	0.000	−0.231
7	4	5	−1.512	−0.645	0.002	−0.118
8	4	14	−2.678	−0.173	0.005	−0.068
9	5	6	−4.714	0.409	0.004	0.004
10	5	8	3.200	1.299	0.009	−0.041
11	6	7	4.252	1.304	0.011	0.040
12	6	11	−3.558	−0.015	0.008	−0.063
13	7	8	1.904	0.424	0.001	−0.068
14	8	9	−0.126	0.072	0.000	−0.405
15	9	39	−0.126	0.477	0.001	−1.213
16	10	11	3.576	0.381	0.005	−0.033
17	10	13	2.924	0.670	0.003	−0.047
18	13	14	2.851	0.282	0.007	−0.117
19	14	15	0.161	0.294	0.000	−0.396
20	15	16	−3.040	−0.839	0.008	−0.103
21	16	17	2.143	0.219	0.003	−0.110
22	16	19	−5.025	−0.357	0.037	0.105
23	16	21	−3.161	0.065	0.007	−0.159
24	16	24	−0.298	−0.986	0.000	−0.070
25	17	18	2.063	0.033	0.003	−0.113
26	17	27	0.077	0.296	0.000	−0.347
27	21	22	−5.909	−0.926	0.026	0.160
28	22	23	0.566	0.616	0.000	−0.203
29	23	24	3.407	−0.052	0.023	−0.046
30	25	26	0.830	−0.476	0.002	−0.535
31	26	27	2.743	−0.047	0.010	−0.159
32	26	28	−1.404	−0.013	0.008	−0.749
33	26	29	−1.902	−0.051	0.020	−0.885
34	28	29	−3.472	0.460	0.016	−0.090
35	2	30	−2.500	−1.765	0.000	0.163
36	31	6	5.412	1.556	0.000	0.673
37	10	32	−6.500	−1.051	0.000	0.880
38	12	11	−0.005	−0.453	0.000	0.008
39	12	13	−0.070	−0.427	0.000	0.008
40	19	20	1.227	1.590	0.003	0.055
41	19	33	−6.289	−2.052	0.031	0.625
42	20	34	−5.056	0.505	0.024	0.489
43	22	35	−6.500	−1.702	0.000	0.592
44	23	36	−5.586	0.025	0.014	0.751
45	25	37	−5.380	2.387	0.020	0.780
46	29	38	−8.245	1.115	0.055	1.068

10.5 Device injections

Table 6: Solved device injections at the accepted equilibrium (per unit on the system base), reconstructed by bus KCL from the stored equilibrium sample and the case network.

Index	Device	Bus	Mode	P pu	Q pu	$\lvert V\rvert$ pu
1	SG31	31	SYNCHRONOUS	5.504	1.602	1.078
2	IBR30	30	GFM	2.500	30.276	1.051
3	IBR32	32	GFM	6.500	26.596	1.022
4	IBR33	33	GFM	6.320	23.047	1.034
5	IBR34	34	GFM	5.080	3.043	0.975
6	IBR35	35	GFM	6.500	27.703	1.071
7	IBR36	36	GFM	5.600	22.824	1.075
8	IBR37	37	GFM	5.400	9.322	0.971
9	IBR38	38	GFM	8.300	20.433	1.003
10	IBR39	39	GFM	10.000	16.568	1.004

11 Event-fence time-domain result

11.1 Per-device injections and the grid-strength diagnostic

At the accepted equilibrium each device's injection is reconstructed by bus KCL from the stored sample and the case network, $S = V\,i^{*}$; the per-device net injections (P_{inj}, Q_{inj}) are the columns of the device-injection table above, unchanged. Expressed per device (per unit on the 100-MVA base): SG31 sends 5.504 pu $+j1.602$ pu; the nine GFM converters send the dispatch P_{inj} = 2.500/6.500/6.320/5.080/6.500/5.600/5.400/8.300/10.000 pu (IBR30, 32–39) and Q_{inj} = 30.276/26.596/23.047/3.043/27.703/22.824/9.322/20.433/16.568 pu. The reactive injections are large because each converter also absorbs its bus's fixed port reactor: $Q_{\text{inj}} \approx Q_{\text{sched}} + Q_r|V|^2$.

The grid-strength screen at the same equilibrium is the **ASSUMED_DIAGNOSTIC** short-circuit ratio

$$\text{SCR}_i = \frac{|V_i|^2}{|Z_{th,ii}|\,S_{\text{rated},i}}, \qquad Z_{th} = Y^{-1}\ \text{(passive pre-fault network)}, \tag{36}$$

where the passive admittance folds the branch series, charging, bus shunts, and the PF-seed load admittance, and S_{rated} is the composition's PROJECT_DERIVED converter rating (the TAMU source leaves no IBR rating). The computation is a column solve of Y at every IBR PCC — the same quantity the production family labels `not_applicable_full_state_source_model` for the *active* SCR gate, published here only as a physical grid-strength diagnostic:

Bus	$\lvert Z_{th}\rvert$ pu	S_{sc} pu	S_{rating} pu	SCR
30	0.019	56.725	44.000	1.289
32	0.022	48.039	44.400	1.082
33	0.023	46.051	36.700	1.255
34	0.043	21.936	11.800	1.859
35	0.021	54.236	43.500	1.247
36	0.026	43.602	35.300	1.235
37	0.030	31.872	18.200	1.751
38	0.029	34.168	35.300	0.968
39	0.024	41.302	33.100	1.248

The computed SCR spans 0.97–1.86 at all nine PCCs — below the conventional strong-grid threshold of 3 at every bus, as expected for a 100-MVA-base island whose only voltage source before the reclose is the GFM fleet itself. The figure is read after the export's load fold and is reported, not gated: the switching supervisor's active gates remain the synchronism triple and the KCL rows, and this diagnostic never enters them.

11.2 Time-domain result in numbers

Table 7 summarizes the measured series of the actual run across the three windows of the chronology (prefault $t < 20$ s, the converter island $20 \le t < 100.499$ s, and the post-reclose $t \ge 100.499$ s), computed on the full 160,000-row record of the C++ run (the per-device series are the bus-KCL reconstruction of Section 1).

Table 7: Measured per-device voltages and island/post-reclose powers of the C++ run, computed on the full record (the ± column is one standard deviation of the post-reclose window).

	Prefault $\lvert V\rvert$ (pu)	Island $\lvert V\rvert$ (pu)		Post-reclose P (MW)	
Device	mean	min	max	mean	±
SG31	1.078	0.983	1.045	544.551	50.345
IBR30	1.051	1.032	1.068	195.392	6.758
IBR32	1.022	0.982	1.040	633.494	10.347
IBR33	1.034	1.000	1.068	628.030	5.867
IBR34	0.975	0.940	1.010	536.574	1.869
IBR35	1.071	1.040	1.101	635.629	7.071
IBR36	1.075	1.052	1.099	551.551	5.852
IBR37	0.971	0.951	0.988	560.175	2.767
IBR38	1.003	0.991	1.014	847.988	5.886
IBR39	1.004	0.972	1.017	1038.274	5.350

Observed system behaviour, all from the stored series:

- The record's only frequency signal is the SG31 rotor speed ($f = 60\,\omega$): the prefault mean is 60.0000 Hz; the island runs above nominal with a peak of 63.077 Hz at $t = 22.082$ s and a 40–55 s mean of 61.660 Hz (60.956–60.964 Hz in the fault window), settling to 60.033 Hz at 150–160 s. The MATLAB edition's COI band cannot be restated here: the C++ record exports no COI or IBR frequency series.
- The bus-voltage envelope spans 0.9335–1.1012 p.u.; both extremes sit inside the 60–60.15 s fault window (0.9335 p.u. at bus 20 at $t = 60.000$ s, 1.1012 p.u. at bus 35 at

t = 60.150 s). The record carries no excursion-flag column, so no flagged-sample count is claimed.

- Island device powers settle to their dispatch (P = 2.625/7.016/6.839/5.545/7.018/6.052/5.879/9.017/ p.u. for IBR30, 32–39 during 40–55 s, sum 60.880 p.u., standard deviation below 10^{-11} p.u.) against the stuck relief the dispatch contract declares; the reclose transaction shows the SG31 5.50 $\rightarrow$ 0 $\rightarrow$ 5.45 p.u. transfer at 20 s and 100.499 s(island injection below 1.9×10^{-9} p.u., post-reclose mean 544.6 MW with standard deviation 50.3 MW, peak 648.7 MW, settling to 550.2 MW and 160.1 MVAr at 150–160 s).

11.3 Per-device mode, power, and voltage traces

Figure 3 shows the supervisory state and the per-device active/reactive power over the full chronology 0–160 s: the top strip is each IBR's grid-forming/grid-following mode (all nine run GFM throughout this C++ record — the record carries no per-sample mode column, so the strip is reconstructed from the event log and the guard record and no mode-switching claim is made), the second strip is the SG31 breaker state (the classical plant trips at 20 s and recloses at 100.499 s), and panels (a)–(b) are the per-device power series in per unit on the system base. The plotted series carry every 10th sample of the 1-kHz record (16,001 rows); every quoted number is computed on the full record.

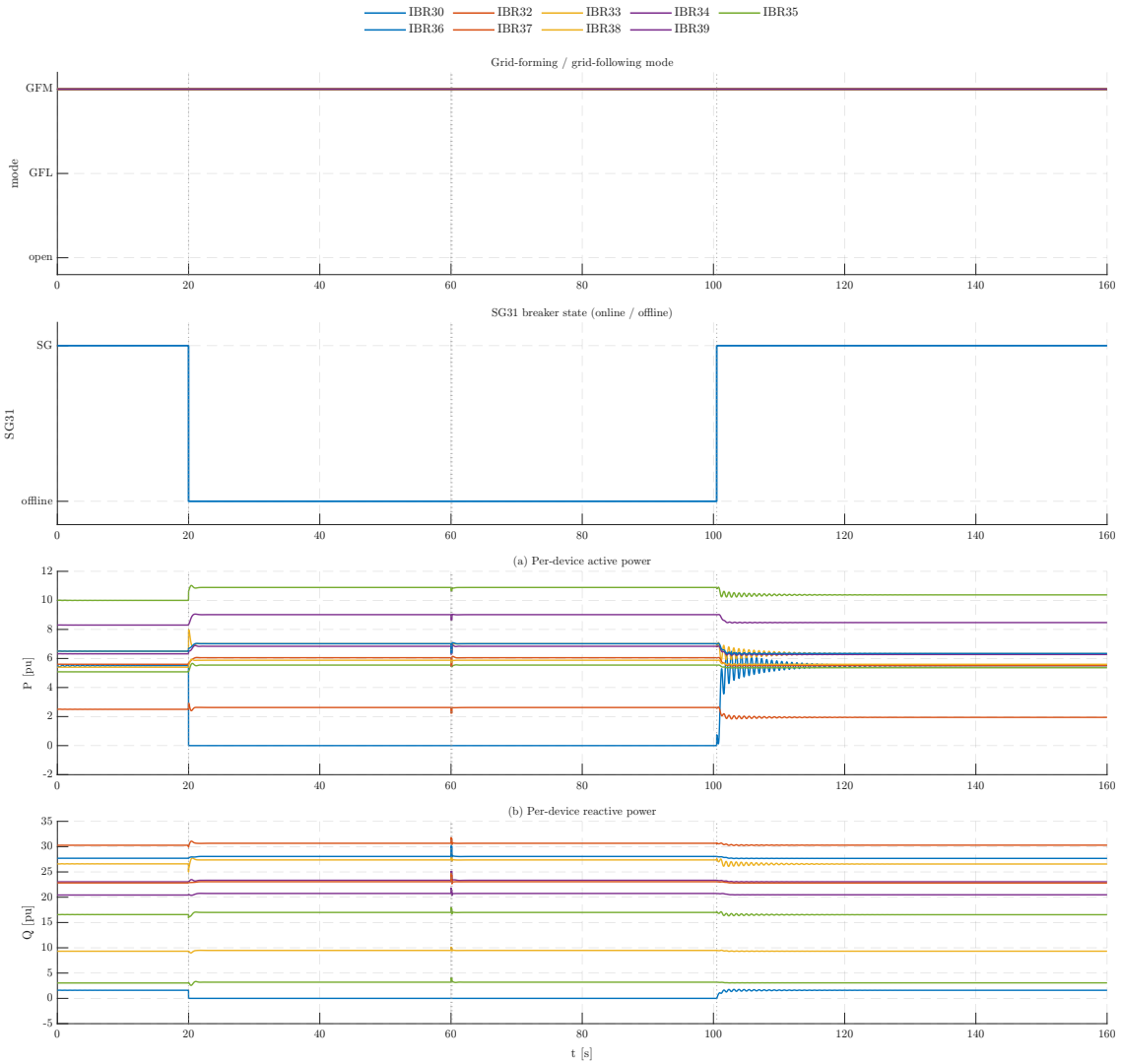


Figure 3: Top strip: grid-forming/grid-following mode of each of the nine IBRs (all nine GFM throughout, reconstructed from the C++ event log and guard record, no display offset). Second strip: SG31 breaker state (SG online, then offline across the island, then SG again at 100.499 s). (a) Per-device active power P_i pu. (b) Per-device reactive power Q_i pu. Per-device series of the C++ run (bus-KCL reconstruction), every 10th sample of the 1-kHz record; $t \in [0, 160]$ s full window.

Figure 4 is the eight-panel electrical response: (a) active power P_i; (b) reactive power Q_i; (c) d-axis current; (d) q-axis current; (e) SG31 rotor frequency; (f) device terminal voltage; (g) network maximum voltage; (h) network minimum voltage — all per unit, all from the C++ run.

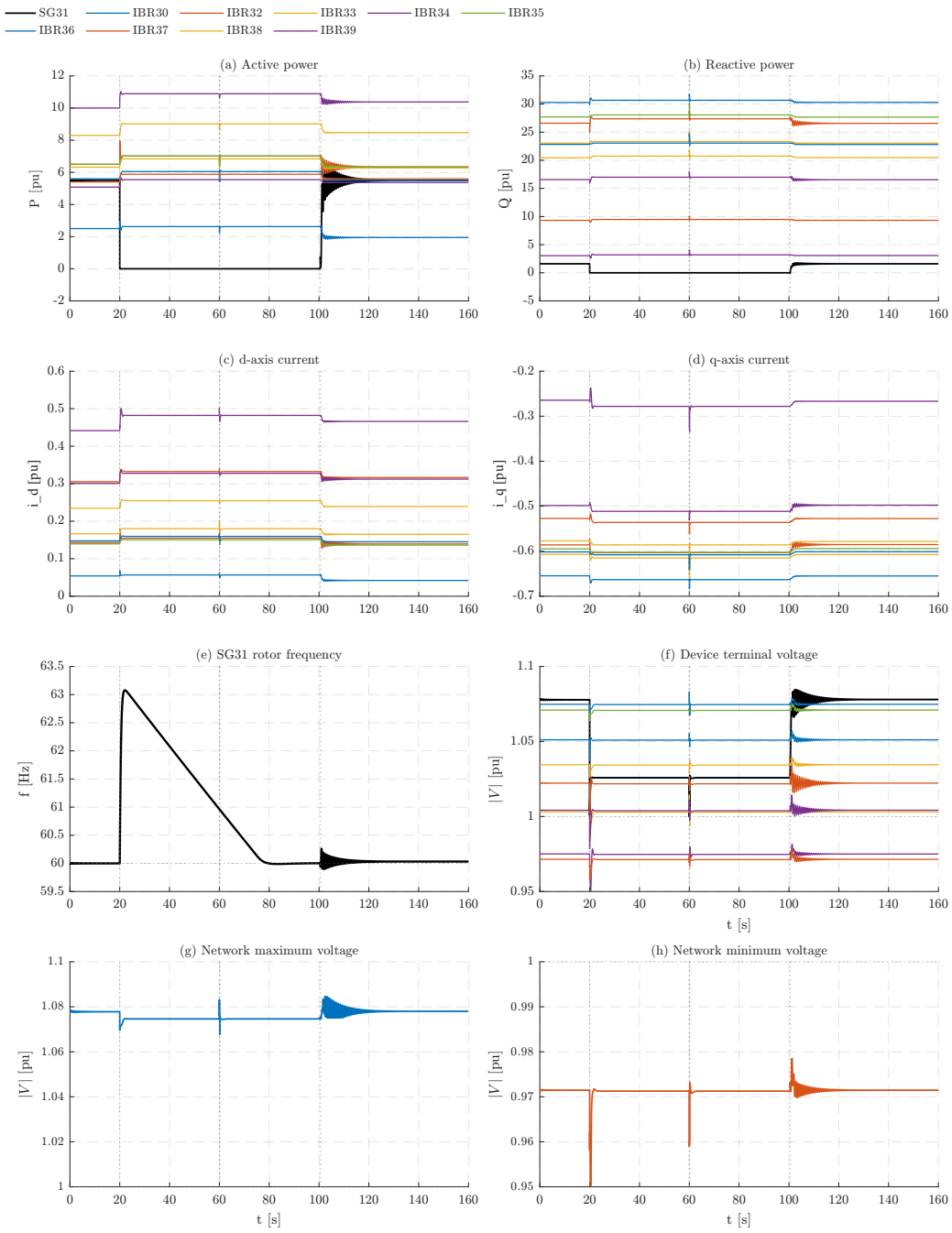


Figure 4: Eight-panel electrical response of the C++ run (bus-KCL reconstruction, every 10th sample of the 1-kHz record, $t \in [0, 160]$ s full window). (a) active power P_i; (b) reactive power Q_i; (c) d-axis current; (d) q-axis current; (e) SG31 rotor frequency ($f = 60\,\omega$) — the record's only frequency signal; no COI or IBR frequency is exported; (f) device terminal voltage; (g) network maximum voltage; (h) network minimum voltage. One common legend at the top of the figure identifies every device trace: SG31 appears in (a), (b), (e) and (f), and (g) and (h) are single network traces. IBR30 blue, IBR32 orange, IBR33 yellow, IBR34 purple, IBR35 green, IBR36–39 cycling the same five colors.

12 Limitations and interpretation

- The run carries no production or hardware certificate: the C++ manifest classifies it `run\_kind=SMOKE` (engine `GridFormLab C++ stage2-ibr`, metadata `ne39\_stage2\_armB\_v1`). No claim in this report is a production or hardware certificate.
- The C++ recorder exports no per-event KCL residuals, no transition-trial evidence, no voltage-excursion flags, and no step-rejection record. Those items of the MATLAB edition are not computable from this artifact; they are reported as not-exported (Table 3) rather than copied from the other edition.
- The per-device power and injection series are a reconstruction (bus KCL through the case network with the load folded at the equilibrium voltage), not a direct export: at equilibrium the route agrees with the aligned dq identity to 3.35×10^{-7} pu, but through the transients the two routes differ by up to 1.24 pu (P) because the aligned-frame v_q is not zero away from the locked equilibrium; the KCL route is the one that satisfies the network at every sample.
- The converter fault contract is balanced positive-sequence RMS with GFM current circles. It is not an EMT, negative-sequence, or zero-voltage model, and it does not reproduce the source GENROU fault-current waveform.
- No kV column is published: the TAMU source leaves bus base kV unspecified, and inventing one would be a fabrication.
- The trip and reclose interval behaviour reflects the seven-state PROJECT_DERIVED reclose plant (six plant states plus the terminal estimator), not a certified external-grid syncheck relay or a GENROU/EXST1/IEEEST machine.
- The reclose guard closed at $t = 100.499$ s with margins $dV = 1.73 \times 10^{-14}$ pu and $df = 5.09 \times 10^{-5}$ pu well inside their thresholds while $d\theta = 4.398°$ sits inside the 10° band — the frozen stage-1 threshold set recorded in `guard\_reclose.json`. The reclose instant therefore follows the guard policy of this engine and differs from the MATLAB edition's 119.55 s (closing provenance table); the difference is a policy difference, not a defect in either run.

13 Conclusion

The report exposes a traceable chain from the TAMU case inputs through the accepted mixed-resource equilibrium, the composite DAE, the event chronology of the `GridFormLab C++ stage2-ibr` engine, and the observed guard-driven SG31 reclose at 100.499 s. The same device equations initialise the equilibrium and are integrated for TS, the network residual is enforced all-row in both, and the final inventory (nine GFM converters and one synchronous SG) is read from the run's committed device configuration — the record carries no per-sample mode column, so no mode-switching claim is made. The plots re-render CSV evidence of the C++ artifact at the report typography, and every quantity the recorder does not export (per-event KCL residuals, transition-trial evidence, excursion flags) remains marked not-exported at the point where the MATLAB edition reported it. The closing provenance table itemises every difference between the two editions, including the reclose-time policy difference.

Data provenance: this C++ run vs. the MATLAB edition

Table 8: Data provenance. Every number in this report traces to the C++ run artifact or to the shared case it consumed; nothing is copied from the MATLAB edition.

Item	This report (C++ `stage2-ibr` run)	MATLAB edition report
Run artifact	`stage2\_smoke\_ne39\_008\_gfm16one_gfl1` (engine `GridFormLab C++ stage2-ibr`, metadata `ne39\_stage2\_armB\_v1`, `run\_kind=SMOKE`, wall 335.6 s)	stored MATLAB run artifact (13 Oct 2026 edition)
Integration	fixed-step implicit trapezoidal, h = 1 ms constant; 160,000 accepted, 0 rejected steps; coupled Newton 300,126 iterations / Jacobian evaluations	adaptive stepper with strict LTE control; 11,522 accepted samples, 147 rejected steps
Reclose instant	guard armed at 100.000 s (`sg\_on`), close committed 100.499 s; margins $dV = 1.73 \times 10^{-14}$ pu, $df = 5.09 \times 10^{-5}$ pu, $d\theta = 4.398°$	strict-transactional close at 119.550 s
Reclose-time difference	guard-policy difference: this run records the frozen stage-1 thresholds (0.05 pu, 0.001 pu, 10°, 0.5 s dwell) in `guard\_reclose.json`; not a numerical defect in either run	—
Frequency evidence	SG31 rotor speed only ($f = 60\,\omega$): island 40–55 s mean 61.660 Hz, peak 63.077 Hz at 22.082 s, tail 60.033 Hz	COI frequency 59.9996–60.0300 Hz
Per-device series	reconstructed by bus KCL from recorded voltages/angles + case network (load folded at the equilibrium voltage); equilibrium cross-check vs. aligned dq identity 3.35×10^{-7} pu	exported from the stored run by each device's injection function
Not exported by this recorder	per-event KCL residuals, transition-trial evidence, excursion flags, rejected-step log, per-sample mode column — reported as *n/e* / not-applicable	recorded and reported
Equilibrium	seeded from the case PF point; solver statistics not present in the artifact	Newton solve logged (1 iteration, residual 3.14×10^{-13} pu)
Shared case inputs	bus/branch tables, IBR ratings and P schedules identical in both editions (verified against the engine's own `case.json`: 46 branches, 11 with off-nominal taps 0.993–1.07 used in the reconstruction; the published branch table lists r, x, b_c only)	same
Figures	regenerated from the C++ record (the three `flagship\_cpp` PDFs in the repository are byte-identical placeholders); plots show every 10th sample of the 1-kHz record (16,001 rows), all statistics on the full record	rendered from the stored accepted samples (11,522 rows, nothing decimated)

References

[1] T. Athay, R. Podmore, and S. Virmani, "A Practical Method for the Direct Analysis of Transient Stability," IEEE Transactions on Power Apparatus and Systems, PAS-98(2), 1979; attribution follows the TAMU distribution metadata.

[2] M. A. Pai, *Energy Function Analysis for Power System Stability*, Springer, 1989.

[3] Texas A&M University, *New England IEEE 39-Bus System* distribution, https://electricgrids.engr.tamu.edu/electric-grid-test-cases/new-england-ieee-39-bus-system/.

[4] EECON49 course material, converter block diagram and parameter table, Eqs. (6)–(29), Figs. 1–2.

[5] Source transcription provenance: `docs/project/NE39_TAMU_MIGRATION.md`.